Ch 04: For and While Loops

Application Program Development

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Introduction to Repetition Structures

- Often have to write code that performs the same task multiple times
  - Disadvantages to duplicating code
    - Makes program large
    - Time consuming
    - May need to be corrected in many places
    - DRY Don't Repeat Yourself
- **Repetition structure**: makes computer repeat block of code as necessary
  - **Condition-controlled loop** uses a `True` / `False` condition to control the number of times that it repeats
  - **Count-controlled loop** repeats a specific number of times

Repetition Structures

A repetition structure causes a statement or set of statements (code block) to execute repeatedly.



The `while` Loop: Condition-Controlled Loops (1 of 2)

- `while` loop: while condition is true, do something
  - Two parts
    1. Condition tested to see if `True` or `False`
    2. Statements (code block) repeated as long as condition is `True`

```
while condition:
    statement
    statement
    etc.
```

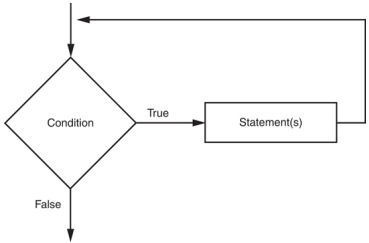


Figure 1: The logic of a while loop (Gaddis, 2021, pg.193)

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The `while` Loop: Condition-Controlled Loops (2 of 2)

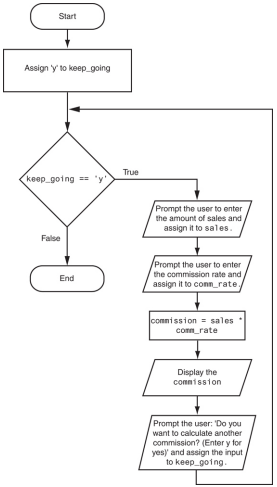


Figure 2: Flowchart for sales commission program (Gaddis, 2021, pg.196)

- In order for a loop to stop executing, something has to happen inside the loop to make the condition `False`
- **Iteration**: one execution of the body of a loop
- `while` loop is known as a **pretest** loop
  - Tests condition performing an iteration
    - Will never execute if condition is `False` at start
    - Required performing some steps prior to the loop

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Infinite Loops

Infinite Loop

Loop that does not have a way of stopping

- Repeats until program is interrupted
- Occurs when programmer forgets to include stopping code in the loop
- **Bug**, use `ctrl-c` (usually) to interrupt running program

- In all but rare cases, loops **MUST** contain within themselves a way to terminate
- Something inside a `while` loop must eventually make the test condition `False`



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The `for` Loop: a Count-Controlled Loop (1 of 2)

- **Count-Controlled loop**: iterates a specific number of times
  - Use a `for` statement to write count-controlled loop
  - Designed to work with a sequence of data itemes
    - Iterates once for each item in the sequence
  - **Target variable**: the variable which is the target of the assignment at the beginning of each iteration

```
for variable in [value1, value2, etc.]:
    statement
    statement
    etc.
```



The `for` Loop: a Count-Controlled Loop (2 of 2)

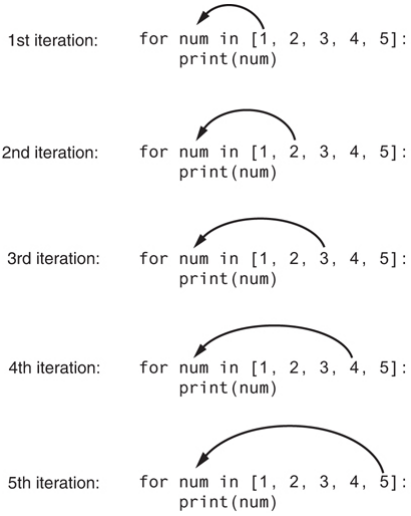


Figure 3: The for loop (Gaddis, 2021, pg.201)

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Using the `range()` Function with the `for` Loop

- The `range()` function simplifies the process of writing a `for` loop
  - `range()` returns an iterable object
    - **Iterable**: contains a sequence of values that can be iterated over
- `range()` characteristics
  - One argument: used as ending limit
  - Two arguments: starting value and ending limit
  - Three arguments: third argument is step value

```
>>> for idx in range(10):
...     print(idx, end=',')
...
0,1,2,3,4,5,6,7,8,9,

>>> for idx in range(5,10):
...     print(idx, end=',')
...
5,6,7,8,9,

>>> for idx in range(2, 10, 2):
...     print(idx, end=',')
...
2,4,6,8,>>>

>>> for idx in range(10, 1, -1):
...     print(idx, end=',')
...
10,9,8,7,6,5,4,3,2,
```



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Using the Target Variable Inside the Loop

- Purpose of target variable is to reference each item in a sequence as the loop iterates
- Target variable can be used in calculations or tasks in the body of the loop
  - Example: calculate square root of each number in a range

```
>>> for number in range(1, 11):
...     square = number**2
...     print(f'{number:5d}{square:5d}')
...
    1    1
    2    4
    3    9
    4   16
    5   25
    6   36
    7   49
    8   64
    9   81
   10  100
```

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Summary

- Repetition structures allow you to write code that performs a block of statements multiple times
  - Avoids code duplication
  - Used to iterate over items in a list or array
- `while` loops typically used for **condition-controlled loops**, where they repeat until some condition becomes `False`
- `for` loops typically used for **count-controlled loops** that iterate a specific number of times, e.g. one time for each item in a list of items
- Use the Python built-in `range()` function to generate explicit iterable list of integers for a count-controlled loop



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References

Gaddis, T. (2021). *Starting out with Python* (5th). Pearson.

